



11861 - Resolving the interplay between massive star winds and hard ionizing radiation with strong lensing at Cosmic Noon

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
CSWA13				
	1	IFU	NIRSpec IFU Spectroscopy	(1) CSWA13

ABSTRACT

The growing population of $z > 6$ galaxies with detections of perplexing emission in highly-ionized gas has highlighted glaring deficiencies in our understanding of massive star populations in the early Universe. Spatially resolved observations of gravitationally lensed arcs at intermediate redshifts represent a critical opportunity to dissect massive stars and their resolved feedback over wavelengths and at a fidelity and spatial resolution inaccessible at higher redshift. And moreover, some Cosmic Noon galaxies host stellar populations closer in scale and environment to those at Cosmic Dawn than can be found in the local Universe. Here we propose deep NIRSpec/IFU spectroscopy probing the rest-optical of a bright lensed galaxy with a remarkable combination of signatures of intermixed highly-ionized gas and hot stellar winds. These observations will map out these gas and massive star tracers together, providing unique insight into the ionizing agents responsible for similar nebular emission at higher- z . The rest-optical data will also constrain unique stellar wind signatures only available in the optical that can diagnose the detailed physics shaping the massive stars in this galaxy, and which have only recently been constrained for the first time in one other galaxy at a similar redshift. Confronting our models for stars and their feedback with these observations will provide insight crucial to correctly interpreting and modeling nebular signatures of ionization and enrichment out to Cosmic Dawn.

OBSERVING DESCRIPTION

Our program will spatially resolve the continuum features in a bright, strongly lensed star-forming galaxy with unique FUV spectral signatures of massive stars and highly-ionized gas: CSWA-13. Our science depends on achieving high SNR for this galaxy in the rest-optical continuum across multiple independent spatial elements. To achieve this, we request 21.4 hours science exposure, corresponding to a total charged time of 26.4 hours. This allows us to probe the continuum at $\text{SNR} \sim 4/\text{IFU spaxel}$ (corresponding to $\text{SNR} > 10$ for 4×4 binned spaxels). The total requested exposure time is based upon ETC calculations for an extended flat continuum source rescaled to match the flux observed to ground based observations and spatial extent matching the rest-optical HST imaging. The G140H grating uniquely satisfies our science wavelength and spectral resolution ($R > 2000$) requirements. We additionally request a short exposure in the G140M to fill the detector gap and cover the [O III] 4959,5007 otherwise missed for a substantial fraction of the galaxy.

Our goal is to maximize on-source exposure and SNR, while minimizing overheads. Data volume considerations from the APT necessitate that we

JWST Proposal 11861 (Created: Wednesday, June 24, 2026, 12:01:24PM Eastern Standard Time) - Overview

utilize NRSIRS2 as opposed to NRSIRS2RAPID. We additionally opt for an 8-point dither pattern to obtain high-quality spatial sampling of our IFU observations. We utilize the maximum-recommended 20 Groups/Int for this configuration (corresponding to ~1500s) to maximize SNR. An 8-point dither with 6 Integrations/Exp then yields the requested 19.6 hours; and the short G140M exposure of 2 hours fits alongside this G140H observation within a single visit.

Proposal 11861 - Targets - Resolving the interplay between massive star winds and hard ionizing radiation with strong lensing at Cos...

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	CSWA13	RA: 12 37 36.1376 (189.4005733d) Dec: +55 33 47.32 (55.56314d) Equinox: J2000		
Comments: Category=Galaxy Description=[Einstein rings, High-redshift galaxies, Starburst galaxies] Extended=YES					

Proposal 11861 - Observation 1 - Resolving the interplay between massive star winds and hard ionizing radiation with strong lensing a...

Observation	Proposal 11861, Observation 1: IFU											Wed Jun 24 17:01:24 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpect IFU Spectroscopy											
Diagnos	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(1)	CSWA13	RA: 12 37 36.1376 (189.4005733d) Dec: +55 33 47.32 (55.56314d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Einstein rings, High-redshift galaxies, Starburst galaxies] Extended=YES											
Template	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	CYCLING		SMALL		1		8				
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	G140H/F100LP	NRSIRS2	20	6	false	true	NONE	8	48	70726.939	277763
	2	G140M/F100LP	NRSIRS2	12	1	false	true	NONE	8	8	7119.378	277763